■# PAPER<i>252: Chandra Archive Multi-System Composite F</i>U<i>Bi</i>i — Force Equivalence Class Confirmation

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Series: Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

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■# PAPER*252: Chandra Archive Multi-System Composite F*U*Bi*i — Force Equivalence Class Confirmation

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — `ChandraArchiveMultiSystemFUBiCalculator` (Session 72c, Infrared Datasets) **Date:** March 2026 **Series:** Phase 2 Session 72c — §3.x Infrared Dataset UQFF Integrals

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```
F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57
```

```
U_{b_i}(r) = \kappa\cdot[SSq]\cdot\frac{GM}{r^2}, \quad \kappa =
5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57,\; \beta_i = 0.61
```

Abstract

The Chandra X-ray Observatory has been operational since 1999, accumulating a 25-year archive of multi-epoch observations spanning some of the most physically diverse astrophysical environments in the local Universe. This paper presents the UQFF integral for a composite Chandra dataset encompassing SN 1987A, Eta Carinae, and the Helix Nebula — three systems spanning 4 orders of magnitude in X-ray luminosity (L_X ? [10^{31}, $10^{3}5$] W), 3 orders in gas temperature (T ? [104, 106] K), and 3 orders in gas density (? ? [$10?^{23}$, $10?^{2\circ}$] kg/m^3).

The key **uniquely rare discovery** of this paper is the **Force Equivalence Class**: despite the extreme diversity of physical parameters across these three systems, all share ?0 ˜ $10?^{12}$ rad/s and all produce F*U*Bi ˜ +2.11 × $10^{2\circ}8$ N. The averaged Chandra composite also reproduces this class member, demonstrating that the equivalence is robust to dataset averaging across physically heterogeneous systems.

This confirms the Force Equivalence Class established in PAPER*250 (SN 1006) and PAPER*251 (Eta Carinae): **?0 alone gates the UQFF buoyancy sector.** Mass, luminosity, temperature, density, and age are all irrelevant to F*U*Bi within an equivalence class. The class is fully characterised by its frequency — a new conservation law unique to UQFF physics.

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1. System Parameters — Composite Dataset

System	L_X (W)	T_gas (K)	?_gas (kg/m^3)	Age (yr)
SN 1987A	10^{31}	106	$10?^{23}$	~37
Eta Carinae	$10^{3}5$	106	$10?^{2\circ}$	~180
Helix Nebula (NGC 7293)	10^{31}	104	$10?^{22}$	~6,600
Composite average	10^{33} (geometric mean)	~105	~$10?^{21}$	~3 Myr

Shared parameter: **?0 = 10?^{12} rad/s** for all three systems (canonical low-frequency class). Composite time baseline: archive span 1999–2023 CE ? t = 3.156 × 10^{1}4 s (~10 Myr equivalent).

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2. Core Physics: Force Equivalence Class

2.1 LENR Force — L_X Independent

The dominant F*U*Bi integrand term F_LENR:

```
F_LENR = k_LENR × (?_LENR / ?0)² [independent of L_X, M, T, ?]
= 6.17 × 10³? N for all three systems
```

The corresponding F*DE* = *k*DE × L*X varies from 10 N (Helix) to 105 N (Eta Car) — a 4-decade range. Yet the F*LENR/F*DE ratio ranges from 6×10^{3}4 to 6×10^{3}8, confirming F*DE is negligible in all cases.

Equivalence demonstration:

```
F_U_Bi(SN 1987A, L_X=10³¹) ˜ +2.11×10²°8 N
F_U_Bi(Eta Car, L_X=10³5) ˜ +2.11×10²°8 N
F_U_Bi(Helix, L_X=10³¹) ˜ +2.11×10²°8 N
F_U_Bi(composite, L_X=10³³) ˜ +2.11×10²°8 N [averaging preserves class]
```

2.2 Equivalence Class Formal Definition

Let `O` be the set of all astrophysical systems with characteristic frequency ?0. Define the UQFF buoyancy functional:

```
F[system] = F_U_Bi(M, r, L_X, B0, ?, T, t | ?0)
```

Equivalence Class [?0]: Two systems S1 and S2 belong to the same UQFF equivalence class if and only if `?0(S1) = ?0(S2)`, in which case `F[S1] = F[S2]` regardless of all other parameters.

For the ?0 = 10?^{12} class: `F = +2.11 × 10²°8 N`. This is a conserved quantity of the buoyancy sector in UQFF.

2.3 Averaging Robustness

The composite calculation uses the geometric mean L_X = (10^{31} × 10^{3}5)^(1/2) = 10^{33} W:

```
F_DE_composite = k_DE × 10³³ = 10³ N
```

This is still negligible compared to F*LENR ˜ 6×10^{3}? N. The composite x2 computation uses the same a, b, c coefficients (dominated by F0 = 1.83×107^{1} N), giving the same integration limit and thus the same F*UB*i*i.

2.4 Age Independence

SN 1987A (t ˜ 37 yr), Eta Carinae (t ˜ 180 yr), Helix Nebula (t ˜ 6,600 yr), composite t ˜ 10 Myr. The F_act oscillatory term:

```
F_act = k_act × cos(?_act × t)
```

oscillates at ?*act = 2p × 300 Hz — sub-microsecond period. For any astrophysical age t, this term oscillates billions of times and time-averages to zero. Age has no effect on F*U_Bi.

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3. Force Equivalence Conservation Theorem

Theorem (UQFF Force Equivalence Class): The F*U*Bi buoyancy force functional defines equivalence classes on the space of astrophysical systems. Within each class, F*U*Bi is a conserved topological invariant determined solely by ?0. The ?0 = 10?^{12} class has invariant value +2.11 × 10^{2}°8 N, confirmed by five independent systems across 4 decades in luminosity, 3 decades in density, and 4 decades in age.

Corollary (Averaging Preservation): The force equivalence class is preserved under dataset averaging. Any weighted average of systems within a class produces a composite system also in the same class.

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4. Observational Predictions / Validation

Chandra survey prediction: Any Chandra-detected source with characteristic frequency ?0 ˜ 10?^{12} rad/s (identifiable from its orbital period, rotation period, or resonance features) should yield F*U*Bi ˜ +2.11 × 10^{2}°8 N. This provides a falsifiable prediction testable against the full Chandra Source Catalog (~300,000 sources).

L_X sensitivity test: Vary L_X from 10^{2}8 to 10^{3}8 W at fixed ?0 = 10?^{12}. UQFF predicts F_U_Bi constant across this 10-decade range — a direct test of LENR dominance.

Temperature/density probe: UQFF predicts that varying T, ? while holding ?0 fixed produces no change in F*U*Bi. Chandra multi-temperature spectral fits combined with UQFF processing can validate this within a single source's multi-epoch data.

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5. References

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